Troy Mayer

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EDUCATION

Purdue University West Lafayette, IN
B.S. Mechanical Engineering and B.S. Applied Statistics, Minor in Economics | GPA 3.32/4 May 2028

SKILLS

- **Languages and tools:** C++17, Python, MATLAB, SQL, Linux, Git, CMake, GoogleTest, GitHub Actions CI, Eigen, MuJoCo, FreeRTOS
- **Methods:** error-state EKF and sensor fusion, rigid-body kinematics, PI and feed-forward control, QP force control, least-squares fitting, Monte Carlo, LSTM and XGBoost
- **Hardware:** ESP32-S3, KiCad, LabVIEW, DAQ, CATIA, NX, Fusion 360, Creo, FEA

ROBOTICS EXPERIENCE

Boston Dynamics Waltham, MA
Sensors and Perception Co-Op, Atlas Jul 2026 – Present

- Computing limb position and velocity at impact with floating-base forward kinematics (quaternions, Rodrigues' rotations, rigid-body velocity composition) from joint encoders, and mapping impacts onto the robot's mesh.
- Estimating impact energy from Khatib's operational-space effective mass and an OLS velocity fit, and building a physics-based MuJoCo impact simulation.
- Built an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM covering ~28k impact events from ~900 robot test logs, with ETL from a time-series telemetry store into SQLite, used by engineers during testing.
- Modeling hardware thermal behavior in Python, fitting heating curves by multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Legged Robot State Estimation and Control | *Personal Project* Aug 2026 – Oct 2026

- Built a contact-aided error-state EKF in C++/Eigen fusing a 1 kHz IMU with leg kinematics on a simulated Unitree Go1: 0.013 m/s velocity RMSE over a 60 s walk vs. 1.38 m/s IMU-only, at ~20 µs per update, Jacobians verified in GoogleTest.
- Ran walking and 500 Hz QP force-balance controllers on the estimate with 0 falls in 5 × 60 s walks, after tracing a 4° lean to weak tilt observability and removing it with a 4 s startup turn.

Active Orthosis, Purdue ASME Biomechanics | *Electronics Team Lead* Apr 2024 – May 2026

- Designed and implemented a 1 kHz PI velocity controller in C++ on an ESP32-S3, tuned from a first-order motor model with feed-forward and anti-windup, protected by 10 fault monitors, a hardware watchdog, and a kill switch.
- Cut EMG-to-motion latency from 103 ms to 43 ms with a Bayesian EMG estimator, validated in closed-loop simulation on real recorded EMG with 26 GoogleTest tests in CI.
- Led a 5-engineer team and redesigned the 4-channel EMG board in KiCad (INA333 front end, right-leg drive, ISO7763 isolation), raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot | *Independent Research, Purdue University & UC3M (Madrid)* Oct 2026 – Present

- Designing push-off control optimized to hit a target launch speed, knee impedance control for landing, and reaction-wheel control to reorient mid-air, with MuJoCo simulation comparing risk-limited hop rules and a pendulum rig to test landings.

ADDITIONAL EXPERIENCE

Caterpillar (via Purdue Data Mine) West Lafayette, IN
Machine Learning Researcher Jan 2026 – May 2026

- Built LSTM and XGBoost models forecasting next-day paint consumption from five years of data with Monte Carlo confidence intervals (4.7% less material used), plus Python pipelines contributing to a 3.3% cut in cycle time.

Chortos Lab, Soft Adaptive Systems and Structures West Lafayette, IN
Undergraduate Researcher Aug 2025 – May 2026

- Wrote a Python computer-vision tracker resolving displacement to 0.01 mm and LabVIEW/MATLAB data acquisition for 50,000+ point datasets, guiding gel formulations that increased force output by over 50%.

Emerson, Discrete Automation (ASCO) Florham Park, NJ
Mechanical and Manufacturing Engineer Intern Jun 2025 – Aug 2025

- Designed and CNC-machined an adjustable spring canister for an industrial rotary valve in Creo with stress calculations and GD&T drawings; ran UL 429 HiPot, torque, bending, and impact validation on solenoid valves.